Tianjian Qin

Computational Biologist & Data Scientist

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Research Profile

Computational scientist specializing in machine learning, probabilistic modeling, and high-performance computing. I build scalable, reproducible pipelines for inference, prediction, and simulation, with a strong interest in applying these methods to biological and environmental observation systems.

- **Machine learning and statistical modeling** for pattern discovery, prediction, anomaly detection, and model comparison;
- **Spatiotemporal and network modeling** for dynamic contact systems and structured observations;
- **Simulation, calibration, and uncertainty quantification** for robust inference from noisy data;
- **Multisource / multimodal data integration** across biological, spatial, temporal, and metadata layers;
- **High-performance computing and reproducible pipelines** for large-scale analysis, simulation, and deployment;
- **Open scientific software and interactive tools** for transparent workflows and cross-disciplinary collaboration.

Education

PhD in Evolutionary Life Sciences (Theoretical Biology) *2019 – 2026*

University of Groningen, Groningen Institute for Evolutionary Life Sciences, Netherlands

Supervisors: *Prof. Dr. Rampal Etienne, Dr. Luis Valente & Dr. Koen van Benthem*

Research on stochastic diversification processes and neural network–based inference from phylogenetic trees. Development of the EVE model (evolutionary relatedness–dependent diversification) and simulation/inference pipelines in R, C++, and Python.

MSc in Ecology (Wetland Ecology and Invasion Biology) *2016 – 2019*

Beijing Forestry University, School of Nature Conservation, China

Supervisors: *Prof. Dr. Hongli Li, Prof. Dr. Feihai Yu*

Research on wetland and aquatic ecological communities, invasion biology, and quantitative ecology. Combined phylogenetic metrics, community composition, trait information, and spatial/environmental data to study resistance to invasion and community assembly across large geographic gradients.

BSc in Biology (Marine Biology) *2012 – 2016*

Nanjing Normal University, School of Life Sciences, China

Broad foundation in marine biology, ecology, and evolution, with training in marine biodiversity, aquatic ecosystems, field sampling, quantitative ecology, and early programming/data analysis. This training established my long-term interest in environmental systems and data-driven biological research.

Research Experience

Post-Doctoral Researcher – Emerging Work in Complex Systems *2025 – present*

Wageningen University & Research, Infectious Disease Epidemiology Group (IDE)

- Designed reproducible pipelines for high-volume spatiotemporal observations, combining data validation/QC, feature engineering, predictive modeling, and uncertainty-aware evaluation on HPC systems.
- Developed physics-informed generative models to synthesize realistic dynamic networks from observational records.
- Developed machine learning models for spatiotemporal prediction, including neural models for farm spatial point patterns and `Transformer`-based link forecasting in dynamic trade networks.
- Integrated heterogeneous data layers (movement records, metadata, spatial information, temporal covariates, and epidemiological context) for multi-source modeling and analysis.
- Developed robust analysis software in `Python`/`R` for network statistics, simulation, and visualization, with emphasis on reusability.
- Built interactive tools (`HerdLink.nl`) enabling domain experts to explore complex dynamic patterns and decision-relevant metrics.
- Worked in HPC/Linux environments for large-scale computation, batch workflows, and model evaluation.

Doctoral Researcher – Theoretical & Computational Biology *2019 – 2026*

University of Groningen, Theoretical Research in Evolutionary Life Sciences Group (TRÊS)

- Developed novel stochastic process models (`EVE`) and simulation algorithms for complex biological systems, with emphasis on identifiability, inference, and interpretability.
- Built high-performance simulation and inference pipelines in `R`, `C++`, and `Python` for large ensembles of structured stochastic outputs.
- Designed deep learning frameworks (`EvoNN`) for parameter estimation and scenario classification from high-dimensional scientific data representations.
- Applied likelihood-based inference, approximate Bayesian computation, and deep learning to evaluate model robustness, redundancy, and uncertainty.
- Developed advanced visualization and summary-statistics workflows for complex structured data (trees/networks/time-varying processes).
- Worked extensively on HPC systems and Linux-based automation for data-intensive computational experiments.
- Released and maintained open-source scientific software with reproducible workflows, documentation, and collaborative development practices.
- Collaborated across biology, statistics, and software engineering on high-performance `R/C++` research packages (`treestats`, `DDD`) and web-based phylogenetic tools (BrassiToL).

Researcher in Ecology & Invasion Biology (MSc) *2016 – 2019*
Beijing Forestry University, Wetland Ecology Group

- Conducted extensive field surveys across 18 provinces in China to quantify wetland plant communities and invasion patterns.
- Developed experience with GIS tools and spatiotemporal modeling by predicting invasive species occurrences at country scale.
- Established automated pipelines of image analysis and phylogenetic tree reconstruction for the research group.
- Analyzed how phylogenetic diversity and species richness affect community resistance to invasion along latitudinal gradient.
- Performed greenhouse experiments manipulating artificial plant communities to study functional and phylogenetic interactions among native species and *A. philoxeroides*.

Teaching and Supervision Experience

Teaching Contributions

- **Teaching assistant** for coding-based assignments and computer practicals in island biogeography yearly courses at the *University of Groningen*.
- **Web application** for teaching and understanding the livestock movement networks.
- **Web applications** for teaching and understanding the temporal evolution of phylodynamics in graphic and spatial contexts.
- **Guest lectures**, **tutorials**, **seminars** at the *University of Groningen* in areas including:
  - stochastic processes in biology;
  - macroevolution and diversification models;
  - deep learning and neural networks;
  - network modeling and statistics.
- **Pair programming sessions** at the *University of Groningen* for test–driven software development and collaborative game design.
- **Lectures, Seminars, Workshops** at *Wageningen University & Research* and *Wageningen Biovetterinary Research* for phylodynamics, birth-death models, and network modeling.

Supervision and Mentoring

- **Informal co-supervision and mentoring of BSc and junior MSc students** at *Beijing Forestry University* on projects involving biostatistics, data visualization, and thesis and paper review.
- **Informal instruction of MSc students** as a PhD candidate at the *University of Groningen*, including guidance on modeling and inference.
- **Informal mentoring of junior PhD candidates** at the *University of Groningen* on research planning, scientific writing, and day-to-day life as an early-career researcher.

Awards, Grants, and Scholarships

Awards

- **Denise Kirschner Best Student Paper Prize**, Journal of Theoretical Biology (2025)
  Inaugural best graduate student paper prize for: *Qin, T., Valente, L., & Etienne, R.S. (2025).*

- **Graduate Academic Innovation Award**, Beijing Forestry University (2018)
  Awarded for innovative graduate research.

Grants

- **Implementatie-impuls COVID-19 programma (ZonMw)** (2026)
  Co-applicant. Project: *SSS-mod Paraatheidspakket bij toekomstige respiratoire uitbraken.*

- **Small Compute Applications (NWO)** (2025)
  Received to support high-performance computing for data-intensive modeling work.

Scholarships

- **CSC–RUG Joint Scholarship** (2019 – 2023)
  Competitive scholarship to support PhD research in theoretical biology.

- **First-Class Academic Performance Scholarship (Graduate)** (2018 – 2019)
  In recognition of outstanding performance within the graduate cohort.

- **First-Class Academic Performance Scholarship (Graduate)** (2017 – 2018)
  In recognition of outstanding performance within the graduate cohort.

Academic Service

Peer Review and Editorial Activity

- **Reviewer** for *Systematic Biology.*

- **Reviewer** for *BMC Public Health.*

- **Reviewer** for *BMC Plant Biology.*

- **Reviewer** for *PLOS Computational Biology.*

- **Reviewer** for *Epidemics.*

- Reviewed manuscripts on topics including infectious disease dynamics, network science, macroevolution, ecological modeling and deep learning.

Community Service

- **Vice President**, Graduate Student Council, School of Nature Conservation, Beijing Forestry University, *2017–2018.*

- **Student representative**, Wetland Ecology Group, School of Nature Conservation, Beijing Forestry University, *2016–2019.*

- **Organizing member**, Overseas Study Association, Nanjing Normal University, *2013–2015.*

- **Registered volunteer**, Xinghuo Volunteer Association, Nanjing Normal University, *2012–2015.*

Professional Skills

Modeling, AI, and Statistical Inference

- Machine learning and GPU deep learning stack for scientific data analysis and prediction.
- Statistical modeling, stochastic processes, and simulation-based inference.
- Uncertainty quantification, model comparison, identifiability, and robustness analysis.
- Dynamic network / graph-based modeling and relational data analysis.
- Spatiotemporal statistics, geospatial analysis, and spatial point pattern modeling.
- Multimodal / multi-source data fusion (biological, spatial, temporal, and metadata layers).

Programming, Data, and Computing Infrastructure

- **Python**: ML/AI pipelines, data processing, graph/network analysis, scientific computing.
- **R**: statistical modeling, package development, simulation, analysis, visualization.
- **C/C++**: high-performance simulation engines and package backends.
- **SQL**: actively developing proficiency for scalable querying, data transformation, and workflows.
- **Spark**: actively learning distributed data processing concepts and workflows for large-scale datasets.
- **Containers**: Docker for dependency management and portable deployment.
- **Linux/HPC**: bash scripting, job scheduling, compute environment and resource management.
- **Git & CI/CD**: collaborative version control, GitHub/GitLab workflows, automated testing/pipelines.
- **Web**: HTML/CSS/JavaScript/D3.js for interactive scientific dashboards and exploratory tools.
- **LaTeX**, **TikZ**, and **PGFPlots** for technical writing and publication-quality figures.

International and Interdisciplinary Experience

- Traveled to 43 countries, gaining first-hand experience of diverse ecosystems and cultures.
- Research experience spanning marine biology, wetland/aquatic ecology, evolutionary biology, and infectious disease epidemiology.
- Proven ability to transfer core computational methods (ML, statistical modeling, simulation, network analysis) across domains with different data structures and scientific questions.
- Strong motivation to collaborate with specialists, technicians, and domain scientists to translate data into robust scientific insight.

Languages

- **Chinese**: native
- **English**: proficient (IELTS Certificate C1, 2019).
- **Dutch**: intermediate (DUO *inburgeringsdiploma* A2, 2025).
- **German**: reading and basic communication (Summer Camp A2.2, 2013).

Publications

Peer-Reviewed Articles

1. **Qin, T.**, van Benthem, K., Valente, L.†, & Etienne, R.† (2025). Parameter estimation from phylogenetic trees using neural networks and ensemble learning. *Systematic Biology.*

2. **Qin, T.**, Valente, L.†, & Etienne, R.† (2025). Impact of evolutionary relatedness on species diversification and tree shape. *Journal of Theoretical Biology.*

3. Sun, K., Liu, X.-S., **Qin, T.-J.**, Jiang, F., Cai, J.-F., Shen, Y.-L., A, S.-H., & Li, H.-L. (2021). Relative abundance of invasive plants more effectively explains the response of wetland communities to different invasion degrees than phylogenetic evenness. *Journal of Plant Ecology.*

4. **Qin, T.**, Zhou, J., Sun, Y., Müller-Schärer, H., Luo, F., Dong, B., Li, H., & Yu, F.-H. (2020). Phylogenetic diversity is a better predictor of wetland community resistance to *Alternanthera philoxeroides* invasion than species richness. *Plant Biology.*

5. Wan, J.-Z., Wang, M.-Z., **Qin, T.**, Bu, X.-Q., Li, H.-L., & Yu, F.-H. (2019). Spatial environmental heterogeneity may be the driver of functional trait variation in *Hydrocotyle vulgaris* (Araliaceae), an aquatic plant invader. *Aquatic Biology.*

6. **Qin, T.-J.*** , Guan, Y.-T.* , Quan, H., Dong, B.-C., Luo, F.-L., Zhang, M.-X., Li, H.-L., & Yu, F.-H. (2019). Growth traits of the exotic plant *Hydrocotyle vulgaris* and the evenness of resident plant communities are mediated by community age, not species diversity. *Weed Research.*

7. **Qin, T.-J.**, Guan, Y.-T., Zhang, M.-X., Li, H.-L., & Yu, F.-H. (2018). Sediment type and nitrogen deposition affect the relationship between *Alternanthera philoxeroides* and experimental wetland plant communities. *Marine and Freshwater Research.*

8. Liu, L., Guan, Y.-T., **Qin, T.-J.**, Wang, Y.-Y., Li, H.-L., & Zhi, Y.-B. (2018). Effects of water regime on the growth of the submerged macrophyte *Ceratophyllum demersum* at different densities. *Journal of Freshwater Ecology.*

Preprints and Under Review

- **Qin, T.**, van Benthem, K., Valente, L.†, & Etienne, R.† Identifying evolutionary relatedness effects on diversification from phylogenies using neural networks. (2026). *bioRxiv.*

Manuscripts in Preparation

- **Qin, T.**, Atamer Balkan, B., Schmid, B., & ten Bosch, Q. Generating livestock movement networks that preserve epidemic behavior with physics-informed machine learning. Manuscript in preparation.

** indicates authors who contributed equally to this work; † indicates joint senior authors.*

Research and Teaching Software

GitHub Public Repositories

- **HerdLink.nl** – Online interactive analyzer for livestock trade networks (`Web/Electron`).
  Role: Developer. A web application offering interactive visualizations and key network statistics at different temporal resolutions of the livestock trade networks of the Netherlands.
  Link: https://HerdLink.nl

- **NetForge** – Livestock trade network models and farm metadata synthesizers (`Python`).
  Role: Developer. Command-line integrations of the `python` scripts for dynamic movement network modeling, generation, evaluation, and transmission simulation.
  Link: https://github.com/EvoLandEco/NetForge

- **NetSpectra** – Higher-order statistics for hybrid dynamic movement networks. (`JavaScript`).

Role: Developer. An interactive simulator and analysis scaffold for exploring hybrid movement networks with temporal edge/node appearance, random-walk mass, and directed spectral signatures.
Link: https://github.com/EvoLandEco/NetSpectra

- **EvoLab** – Watching diversification before it gets flattened into a final tree. (`JavaScript`).
  Role: Developer. An interactive diversification simulator for lineage history, coarse-grained taxa, and trait-based tree reconstruction.
  Link: https://github.com/EvoLandEco/EvoLab

- **evesim** – Evolutionary relatedness–dependent diversification simulator (`R/C++`).
  Role: Lead developer. Tools for simulating EVE birth–death processes and exploring how evolutionary relatedness influences diversification.
  Link: https://github.com/EvoLandEco/evesim

- **EvoNN** – Neural network–based parameter estimation for phylogenetic trees (`R/Python`).
  Role: Developer. Framework for parameter inference and scenario classification from phylogenetic trees using pre-trained neural networks.
  Link: https://github.com/EvoLandEco/EvoNN

- **miniape** – A `JavaScript`-based partial port of the popular R package `ape`
  Role: Developer. Enabling basic manipulation of the `phylo` object on web applications.
  Link: https://github.com/EvoLandEco/miniape

- **treestats** – Fast summary statistics for phylogenies (`R/C++`).
  Role: Collaborator. High-performance implementation of phylogenetic statistics for large-scale tree analyses.
  Link: https://github.com/thijsjanzen/treestats

- **DDD** – Diversity-Dependent Diversification simulation and inference (`R/C++`).
  Role: Collaborator. Implements maximum likelihood and bootstrap methods based on the diversity-dependent birth-death process to test whether speciation or extinction are diversity-dependent, under various models including various types of key innovations.
  Link: https://github.com/rsetienne/DDD

- **DAISIE** – Dynamic Assembly of Islands through Speciation, Immigration and Extinction (`R/C++`).
  Role: Collaborator. Simulates and computes the (maximum) likelihood of a dynamical model of island biota assembly through speciation, immigration and extinction processes.
  Link: https://github.com/rsetienne/DAISIE

Online Tutorials

- **Hybrid Dynamic Networks and Higher-Order Statistics.**
  Explore `NetSpectra`, a hybrid COROP–farm network lens that turns livestock movements into higher-order dynamics and spectra.
  Link: https://qtj.me/blog-Hybrid-COROP-Farm.html

- **HerdLink: Exploring NL Livestock Trade as a Network.**
  Livestock trade data looks nicer with dashboarding. This post introduces `HerdLink.nl`, an interactive NL livestock trade network explorer.
  Link: https://qtj.me/blog-herdlink.html

- **Sub-Species Level Diversification and Phylogeny Reconstruction.**
  Diversification is an event-by-event process, and a tree can hide what actually happened. This post walks through a toy simulator, `EvoLab`, turning "tree-like" history into a network.
  Link: https://qtj.me/blog-Subspecies-Level-Diversification.html